

1. Introduction to HTML

HTML stands for **HyperText Markup Language**.

It is the standard language used to create and structure web pages.

HTML defines:

- What content is present on a page
- How the content is structured
- The meaning of text, images, links, tables, and forms

HTML is **not a programming language**.

It is a **markup language** that uses tags to mark content.

2. HTML Tags – Basic Concept

HTML uses **tags** to define elements.

A tag is written using angle brackets `< >`.

Types of Tags

1. Opening tag: `<p>`
2. Closing tag: `</p>`
3. Self-closing tag: `<img>`, `<br>`, `<hr>`

Most HTML elements have both opening and closing tags.

3. HTML Document Structure

Every HTML document must follow a fixed structure.

```
<!DOCTYPE html>
<html>
<head>
  <title>Page Title</title>
</head>
<body>
  Page content
```

```
</body>
</html>
```

This structure is mandatory for all HTML pages.

4. DOCTYPE Declaration

Tag

```
<!DOCTYPE html>
```

Purpose

- Informs the browser that the document uses HTML5
- Prevents rendering issues
- Must be the first line of the HTML file

The DOCTYPE declaration does not have a closing tag.

5. `<html>` Tag

Purpose

- Root element of the HTML document
- Wraps the entire webpage

Syntax

```
<html>
</html>
```

Common Attribute

```
<html lang="en">
```

The `lang` attribute specifies the language of the page.

6. <head> Tag

Purpose

- Contains metadata about the document
- Content inside <head> is not visible on the page

Common Elements Inside <head>

- <title>
 - <meta>
 - CSS and JavaScript references
-

7. <title> Tag

Purpose

- Defines the title of the web page
- Displayed in the browser tab
- Used by search engines

Example

```
<title>HTML Basics</title>
```

8. <meta> Tag

Purpose

- Provides metadata information about the document

Common Meta Tags

```
<meta charset="UTF-8">
```

```
<meta name="viewport" content="width=device-width, initial-scale=1.0">
```

- `charset` defines character encoding
 - `viewport` makes the page responsive on mobile devices
-

9. `<body>` Tag

Purpose

- Contains all visible content of the web page

Example

```
<body>  
  <h1>Hello World</h1>  
</body>
```

10. Heading Tags (`<h1>` to `<h6>`)

Purpose

- Define headings and subheadings
- Create content hierarchy

Example

```
<h1>Main Heading</h1>  
<h2>Section Heading</h2>  
<h3>Subsection Heading</h3>
```

Rules

- Use only one `<h1>` per page
 - Do not use headings for styling purposes
-

11. Paragraph Tag (`<p>`)

Purpose

- Defines a block of text

Example

```
<p>This is a paragraph.</p>
```

Browsers automatically add spacing around paragraphs.

12. Line Break and Horizontal Rule

Line Break (
)

```
Hello<br>World
```

- Breaks text into a new line
- Self-closing tag

Horizontal Rule (<hr>)

```
<hr>
```

- Creates a horizontal divider
-

13. Text Formatting Tags

Bold and Importance

```
<b>Bold text</b>
```

```
<strong>Important text</strong>
```

Italic and Emphasis

```
<i>Italic text</i>
```

```
<em>Emphasized text</em>
```

Other Formatting Tags

```
<u>Underlined text</u>
```

```
<small>Small text</small>
```

```
<mark>Highlighted text</mark>
```

Semantic tags like `<strong>` and `<em>` provide meaning and improve accessibility.

14. Anchor Tag (`<a>`)

Purpose

- Creates hyperlinks

Example

```
<a href="page.html">Open Page</a>
```

Common Attributes

- `href`: Destination
- `target`: Controls where the link opens

```
<a href="page.html" target="_blank">Open in new tab</a>
```

15. Image Tag (`<img>`)

Purpose

- Displays images on the page

Example

```

```

Attributes

- `src`: Image path
- `alt`: Alternative text
- `width`, `height`: Size control

The `<img>` tag does not have a closing tag.

16. List Tags

Unordered List

```
<ul>
  <li>Item One</li>
  <li>Item Two</li>
</ul>
```

Ordered List

```
<ol>
  <li>Step One</li>
  <li>Step Two</li>
</ol>
```

Description List

```
<dl>
  <dt>HTML</dt>
  <dd>Markup language</dd>
</dl>
```

17. Table Tags

Example

```
<table border="1">
  <tr>
    <th>Name</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Praveen</td>
    <td>25</td>
  </tr>
</table>
```

Table Elements

- `<table>`: Table container
- `<tr>`: Table row

- `<th>`: Header cell
 - `<td>`: Data cell
-

18. Form Tags

Form Container

```
<form method="post">  
</form>
```

Input Fields

```
<input type="text">  
<input type="email">  
<input type="password">  
<input type="number">  
<input type="date">
```

Label

```
<label>Name</label>  
<input type="text">
```

Textarea

```
<textarea rows="4"></textarea>
```

Button

```
<button type="submit">Submit</button>
```

Forms are used to collect user input.

19. `div` and `span`

`div`

- Block-level element
- Takes full width

```
<div>Content</div>
```

span

- Inline element
- Takes only required space

```
<span>Text</span>
```

20. Semantic HTML Tags

Semantic tags give meaning to page structure.

Tag	Purpose
<code><header></code>	Top section
<code><nav></code>	Navigation
<code><main></code>	Main content
<code><section></code>	Grouped content
<code><article></code>	Independent content
<code><footer></code>	Bottom section

Example

```
<header>Header</header>
<main>
  <section>
    <article>Article Content</article>
  </section>
```

```
</main>  
<footer>Footer</footer>
```

21. HTML Comments

Syntax

```
<!-- This is a comment -->
```

Comments are not visible in the browser and are used for explanation.